Week 5

Course Level Objectives:

- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Interpret the slope of a regression line in context with correct units. (CLO 7) **2.** Interpret the intercept of a regression line in context and evaluate whether it is meaningful. (CLO 7) **3.** Use a regression equation to predict a response value. (CLO 7) **4.** Compute and interpret a residual. (CLO 7) **5.** Locate positive and negative residuals on a scatterplot. (CLO 7) **6.** Use technology to find the least-squares regression line. (CLO 7) **7.** Explain what “least squares” means. (CLO 7) **8.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, and influential outliers. (CLO 7) **9.** Explain why a strong regression relationship does not by itself establish cause and effect. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Participation: Two Variable Relationships – Tue: correlation review Participation: Regression Line – Thu: baseball game length and commute-time examples Your Turn: Putting It All Together (used-car price vs. age: slope, intercept, prediction, residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM) Take-Home Quiz 3 (due Tue Oct 6, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus